UIC UNIVERSITY OF ILLINOIS AT CHICAGO

COURSE REQUEST SYSTEM - PRODUCTION
UPDATE PENDING REQUEST
***** DRAFT *****

10/13/2016 *Course:* **CHE 451** *Version:* **1** *Title:* **Renewable Energy Technologies** *User Unit Code:* **2224**

Date Entered: 10/13/2016 9:59:18 AM **by** Karen Milla, Customer Service Representative III
Last Modified: 10/13/2016 10:47:50 AM **by** Karen Milla, Customer Service Representative III

General reason for this request:

A course titled "Renewable Energy Technologies" has been offered every Fall semester since 2011 under the general CHE 494 Selected Topics in Chemical Engineering rubric. The course provides senior undergraduate and graduate students fundamentals of renewable energy sources, energy storage and conversion technologies, and promises and challenges of the so-called Hydrogen Economy. The course has been very well received by our students due to increasing interest in renewable energy and sustainability. Thus, the ChE Department wishes to offer the course permanently under its own rubric, number and name.

This request should be processed at the same time as the following related academic program request(s):

None

Requested type of approval and effective date:

Permanent approval effective Fall semester, 2017

Course Subject (Rubric): CHE / Chemical Engineering
Primary Unit: Chemical Engineering
Course Number: 451
Course Version: 1

Course Title: Renewable Energy Technologies
Transcript Title: Renewable Energy Technologies

Course Description:

Fundamentals of renewable energy technologies; solar, wind, biomass. Introduction to energy storage technologies; batteries and fuel cells, and analysis of the hydrogen economy.

Notes to Students:

None

Cross Listings:

None

Previous Course Number and/or Subject:

None

Expected Registration:

Junior and senior undergraduate
Graduate College
Graduate professional

Type of Course:

Selective for the following programs: Proposed course will satisfy technical elective credits towards the bachelors degree in
Chemical Engineering

Course Learning Outcomes:

1) Students will be able to understand the fundamentals and essential concepts used in renewable energy technologies such as solar, wind and biomass.
2) Students will be able to understand and analyze energy storage technologies such as batteries, fuel cells, and supercapacitors.

addressing key ideas, as well as technical and economic issues related to renewable energy.

Course Learning Outcome Assessment Methods:

Semi-Weekly homework assignments will assess students' knowledge of course material and the big picture of local and global renewable energy challenges and opportunities
2) Mid-Term exam will focus on students' ability to conduct a techno-economic feasibility study of a specific renewable energy project
3) A final team project including a written report and group presentation to assess students' ability to identify an idea or problem related to renewable energy and present a unique approach to identify opportunities and address key challenges.

Purpose of this course in relation to overall curriculum and relationship to other courses offered by primary unit:

The course discusses the topic of renewable energy from a chemical engineering viewpoint and which is timely and of interest to students and researchers across various engineering disciplines. The course does not overlap with any other Chemical Engineering courses.

Relationship of this course to similar courses offered by other academic units:

None

Major Topics:

Part I: Sources of Renewable Energy (18 hrs)
1) Solar
2) Wind
3) Biomass
4) Other sources

Part II: Energy Storage and Conversion (9 hrs)
5) Batteries and Fuel Cells

Part III: The Hydrogen Economy (9 hrs)
6) Hydrogen Production (reforming and electrolysis)
7) Hydrogen Storage and Transportation

Part IV: Case Studies (Applications) (9 hrs)
8) Hybrid Wind/Solar Hydrogen Systems
9) Hybrid Desalination Systems: fuel cell/desalination, solar desalination

TOTAL: 45 hrs

Sample Sources and Resource Materials:

"Hybrid Hydrogen Systems for Stationary and Transportation Applications", Said Al-Hallaj and Kristofer Kiszynski, 1st Edition, 2011, Springer Publishing, ISBN 978-1-84628-466-3

Prerequisite(s):

None

Recommended Background:

None

Corequisites:

None

Restrictions:

None

Credit Restrictions:

None

Credit Hours:

3 undergraduate hours; 4 graduate hours.

Type of Instruction:

Type of Instruction	*Contact Hours/Week Over 15-Week Term*
Lecture	3

Grading Mode: Normal

Sign-Offs:

To be forwarded for review, sign-off, and comment to the following administrators:

Chairperson(s) or director(s) of department(s) or unit(s):

None

Dean(s) of college(s) or school(s):

None

Comments related to request for sign-off:

None

Frequency of Offering:

Frequency	*Special Notes*
Fall of every year.	

Special Technology or Equipment Support Request:

None

Faculty Proposer(s):

Said Al-Hallaj

PRIMARY DEPARTMENT APPROVAL STATUS

NOT YET APPROVED by **Chemical Engineering**

PRIMARY COLLEGE APPROVAL STATUS

NOT YET APPROVED by **Engineering**

OFFICE OF VICE CHANCELLOR FOR ACADEMIC AFFAIRS APPROVAL STATUS

NOT YET APPROVED